Finite-size decay-mode hybridization near a mean-field fold in an open Rydberg ensemble

OpenQuantum
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How does finite-size decay-mode mixing near a mean-field fold appear in physical relaxation? We study a uniformly interacting Rydberg ensemble with independent local decay, retaining the complete permutation-invariant operator space. Two real decay eigenvalues exhibit an avoided crossing on the monostable side of the fold, with an approximately exponential decrease of their separation over $N = 24$–56. Physical product-state perturbations identify an exchange of high-branch relaxation residues between modes whose common decay rate remains finite. A biorthogonal reduction identifies the invariant coupling product, while exact coherence elimination recovers both eigenvalues as self-consistent roots of one effective branch. Reconstructing observable residues additionally requires the initial-coherence source and spectral-derivative normalization. Fixed-preparation controls at $N = 24$–48 distinguish this spectral exchange from the complete density transient. Over $0 \le \gamma t \le 12$, the sampled signals remain smooth across the narrow mixing region, despite substantial changes of individual residues. The coherence-to-population peak is smaller than its independent-atom counterpart, although mean field fails to reproduce the complete transient. These results quantify the observable limits of the finite-size spectral structure without deriving an asymptotic switching action or fold correspondence.

I. INTRODUCTION

Driven Rydberg ensembles connect nonlinear mean-field bistability with finite quantum dynamics [1–4]. A stable mean-field branch terminates at a saddle-node fold, whereas a finite quantum system evolves under a linear, trace-preserving Liouvillian. Relating these descriptions requires distinguishing stationary mean-field roots, eigenvalues of their stability Jacobian, and eigenvalues of the many-body generator.

This distinction matters for exceptional-point (EP) interpretations of nonlinear switching and stationary-root singularities [4–6]. A fold of stationary solutions need not be a defective eigenvalue of the mean-field Jacobian. The stationary sector of a finite-dimensional quantum dynamical semigroup cannot contain a nontrivial Jordan block, which would produce unbounded polynomial growth [7]. In the spectral-separation framework, metastability involves low-lying decay modes separated from faster relaxation [8, 9]. More generally, large eigenmode-expansion coefficients can prolong transients, so decay eigenvalues alone do not determine relaxation times [10, 11].

For uniformly interacting spins with local decay, collective switching and steady-state correlations have been studied [1, 2], and an exact finite-N steady state is available [12]. Exponentially slow switching is central to first-order dissipative transitions [13]; quantum-instanton methods predict switching rates for collective spin systems, including local dissipation [14, 15]. A related conjecture connects a symmetry-selected steady-state purification potential to the ordinary Liouvillian gap [16]. These results concern relaxation toward the stationary sector, rather than the separation of two modes with a finite common decay rate.

Finite-size avoided crossings from mode mixing [17] and few-mode descriptions of hysteresis [18] also have precedents. Cabot, Giorgi, and Zambrini connected a Liouvillian EP to a mean-field bifurcation between phase diffusion and quantum activation in a driven oscillator [19]. Collective-spin models exhibit spectral doublets approaching finite nonzero decay rates [20], with preparation-dependent visibility and cancelling spectral residues [21]. These precedents leave a model-specific question for the locally damped Rydberg ensemble: how does a real-mode avoided crossing redistribute response, and does that redistribution produce a sharp density transient?

We address this question at $\Omega = 2\gamma$ and $V = 10\gamma$ in the complete permutation-invariant operator space. We first identify mode character using physical preparations and signed observable residues. Biorthogonal projection and coherence elimination then connect the splitting to its effective roots and response weights. Finally, fixed-preparation scans and independent-atom and mean-field controls test the implications for complete transients. This separates the finite-size spectral structure from claims about observable enhancement or asymptotic switching.

II. MODEL, MEAN-FIELD FOLD, AND FINITE-SIZE SPECTRUM

We consider N two-level atoms with excited-state number $\hat{N}_e = \sum_i \hat{n}_i$, collective spin $\hat{J}_\alpha = \frac{1}{2}\sum_i \hat{\sigma}_i^\alpha$, and independent decay:

$$\dot{\rho} = \mathcal{L}_N \rho = -i[\hat{H}, \rho] + \gamma \sum_i \mathcal{D}[\hat{\sigma}_i^-]\rho, \tag{1}$$

$$\hat{H} = -\Delta \hat{N}_e + \Omega \hat{J}_x + \frac{V}{2N}\hat{N}_e(\hat{N}_e - 1), \tag{2}$$

where $\mathcal{D}[c]\rho = c\rho c^\dagger - \{c^\dagger c, \rho\}/2$. This is the uniformly interacting, locally damped model of Refs. [1, 2], within

the exactly solvable steady-state family of Ref. [12]. The Supplemental Material (SM) gives the finite-N parameter mappings, including the detuning shift associated with $\hat{N}_e(\hat{N}_e-1)$. We set $\gamma=1$, so frequencies and eigenvalues below are measured in units of γ, and time in γ^{-1}.

Permutation invariance reduces the operator space to dimension $\binom{N+3}{3}$ [22]. We retain every Dicke j block and the transitions between blocks caused by local decay. This differs from the maximum-spin effective-jump approximation used in recent Rydberg switching calculations [23]. Throughout, "full generator" denotes this complete permutation-invariant operator space, which contains the product states and observables considered here. It does not enumerate modes in other permutation-symmetry sectors.

In the large-N factorized limit [1, 2], write the single-atom coherence as $\rho_{eg}=u+iv$ and the excited population as n. The mean-field equations are

$$\begin{aligned}\dot{n}&=-n-\Omega v,\\ \dot{u}&=-(\Delta-Vn)v-u/2,\\ \dot{v}&=(\Delta-Vn)u-\Omega(1-2n)/2-v/2.\end{aligned}\tag{3}$$

Their stationary populations satisfy

$$F(n)=n[(\Delta-Vn)^2+1/4+\Omega^2/2]-\Omega^2/4=0.\tag{4}$$

At $\Omega=2$, $V=10$, the left fold is at $(\Delta_f,n_f)\simeq(3.60,0.157)$. Using $F=\partial_n F=0$, the characteristic polynomial of the Bloch Jacobian at a fold reduces to

$$\chi(\lambda)=\lambda[(\lambda+1)^2+\Omega^2/2].\tag{5}$$

Thus the zero eigenvalue is simple, including when the fold reaches the cusp. Li, Sun, and Yi [6] obtained square-root and cube-root singularities of stationary excitation densities near fold and cusp points. Those exponents describe roots of the nonlinear stationary equation; Eq. (5) concerns the distinct linear stability problem. In particular, a triple stationary root does not imply an EP3 of this Jacobian. The coalescing fixed points and the surviving high-population branch must also be distinguished; Eq. (5) describes the former.

We locate a local minimum of the separation between two real decay eigenvalues and denote it by $\Delta_*(N)$. Define

$$g_N=\frac{\lambda_+(\Delta_*)-\lambda_-(\Delta_*)}{2}>0.\tag{6}$$

Across $N=24,28,\ldots,56$, the crossing drifts toward the fold while remaining on its monostable side, and the half-gap decreases by almost two orders of magnitude [Fig. 1]. An exponential fit gives $\sigma\simeq0.14$, with a small systematic drift as the size window changes (SM). This finite-window trend determines neither a limiting exponent nor convergence of $\Delta_*(N)$ to Δ_f.

The small splitting must not be confused with a slow decay rate. At the $N=32$ crossing, $g_N\simeq0.01$ while the mean decay rate is $-\bar{\lambda}\simeq0.75$, where $\bar{\lambda}=(\lambda_++\lambda_-)/2$. An oscillatory pair decays more slowly than both real modes (SM). Thus g_N is an excited-mode separation, distinct from the ordinary Liouvillian gap $\Gamma_{\rm diss}=-\max_{\lambda\neq0}\mathrm{Re}\,\lambda$.

III. PHYSICAL RESPONSE AND MODE CHARACTER

A Hermitian right eigenoperator at a nonzero decay eigenvalue is traceless; its signed excitation-number profile is not a probability distribution. We test mode character through observable residues, whose dependence on both preparation and detection is essential in nonnormal dynamics [10, 21].

Let R_i and W_i be right and transpose-left eigenvectors of the stored generator, normalized by $W_i^TR_j=\delta_{ij}$. The transpose denotes the algebraic dual of the chosen vectorization. For the excitation-density observable $o^T\rho=\langle\hat{N}_e\rangle/N$, an initial difference $\delta\rho$ gives the exact contribution of a simple mode,

$$\delta n_i(t)=a_ie^{\lambda_it},\qquad a_i=(o^TR_i)(W_i^T\delta\rho).\tag{7}$$

Each residue is invariant under reciprocal rescaling of its left and right eigenvectors. These signed weights are not probabilities of physical states or metastable phases.

We use two physical preparations, whose construction is specified in the SM. The first subtracts positive product states displaced slightly in opposite directions along the real relaxation direction of the high mean-field fixed point. This high-branch preparation is reconstructed at each detuning to follow local mode character. The second subtracts the exact stationary state from a fixed product state prepared at the left-fold Bloch vector. It probes low-population response without assuming a finite-size metastable state outside the bistable interval.

Across the crossing, the dominant high-branch response transfers from λ_+ to λ_- [Fig. 2(a)]. The exchange persists in a separate $N=40$ calculation, and reducing the preparation displacement confirms the perturbative response (SM).

The fold-state preparation instead excites both modes, with large opposite residues near the crossing [Fig. 2(b)]. Large opposite residues can cancel in a bounded physical signal, as analyzed for collective-spin exceptional spectral phases [21]. These finite-size values establish neither a $1/g_N$ residue asymptote nor a real-axis Jordan limit; the SM states the additional limiting conditions.

The signed excitation-number profile $P_i(k)=\mathrm{Tr}(\Pi_kR_i)$ provides a complementary diagnostic. Its absolute-profile fraction below a fixed low-density cutoff exchanges between the branches, alongside the high-branch response [Fig. 2(c)]. Neither profile is purely localized there. This supports enhanced low-density character of the second branch, without identifying it as a universal ghost-escape mode.

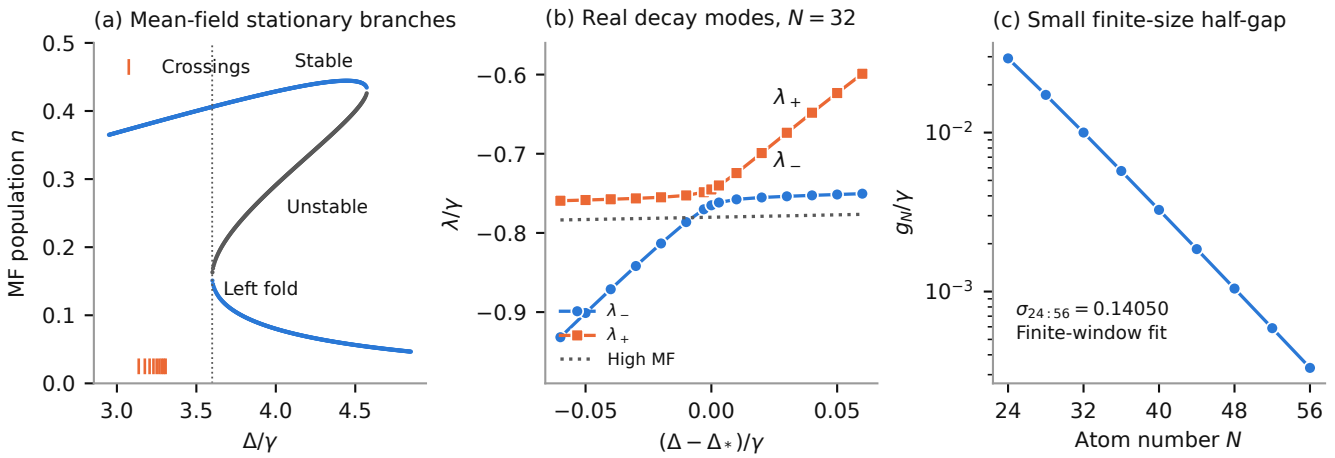


FIG. 1. Mean-field geometry and the finite-size real-mode crossing at $\Omega = 2$, $V = 10$. (a) Stationary populations: stable branches in blue, unstable branch in gray. Orange ticks mark the real-axis crossing positions for $N = 24, 28, \ldots, 56$; all lie on the monostable side of the left fold. (b) The two target real eigenvalues for $N = 32$, ordered as $\lambda_- < \lambda_+$, versus $\delta = \Delta - \Delta_*$. The dotted curve is the real relaxation eigenvalue of the surviving high mean-field branch. The modes exchange high-branch response, but the finite-size rates need not equal the mean-field rate. (c) Half-gap at its local minimum. The quoted exponent is a finite-window fit, not an established large-N limit.

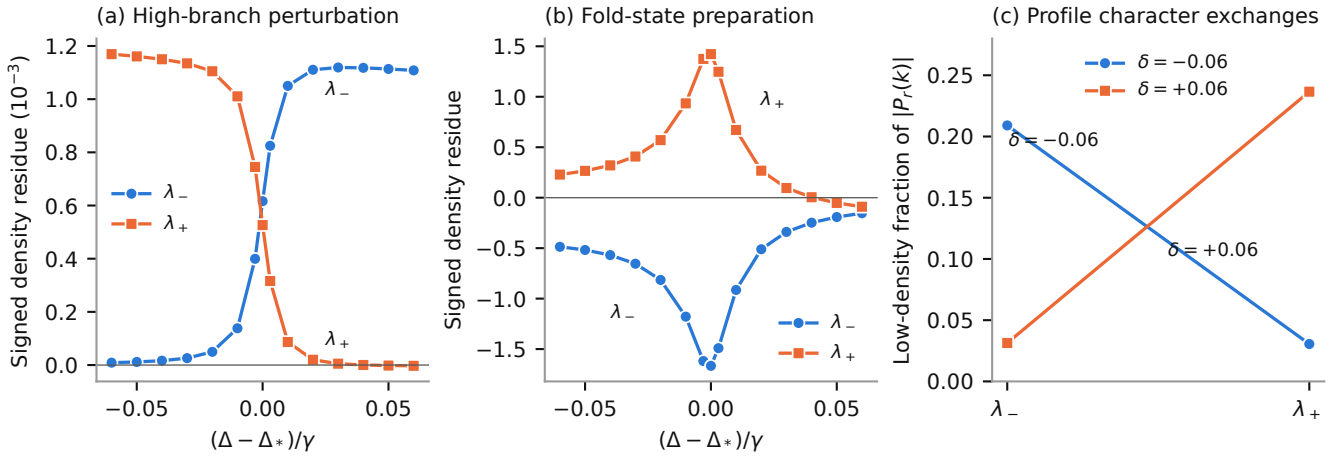


FIG. 2. Physical residues and exchange of mode character, $N = 32$. (a) Residues for product-state differences about the high mean-field fixed point, reconstructed at each detuning. (b) Residues for a fixed fold product state relative to the current stationary state. Large opposite residues cancel in the physical signal. (c) Fraction of the absolute excitation-number profile below $k/N = 0.22$ at the two endpoints. This normalization-independent ratio characterizes a signed eigenoperator, not a population probability of a physical state.

IV. BIORTHOGONAL TWO-MODE REDUCTION AND COMPLEX-DETUNING EPS

A. Local reduction at the gap minimum

Write $\mathcal{L}_N(\Delta) = L_0 + \Delta L_1$, with $L_1\rho = i[\hat{N}_e, \rho]$. At Δ_*, choose bases R, W of the exact two-dimensional right and left invariant subspaces with $W^T R = I$. Diagonalizing $W^T L_1 R$ defines a local diabatic basis with distinct real slopes s_a, s_b. Holding that basis fixed gives

$$H_P(\delta) = \begin{pmatrix} e_a + s_a\delta & V_{ab} \\ V_{ba} & e_b + s_b\delta \end{pmatrix}, \qquad \delta = \Delta - \Delta_*. \quad (8)$$

Here a denotes the smaller-absolute-slope branch. At $\delta = 0$ the projection is exact. Coupling to the complementary invariant subspace produces a spectral self-energy starting at order δ^2 (SM); its small absolute size does not alone establish its smallness relative to an exponentially small coupling.

The usual two-by-two discriminant [24] gives the

squared separation and its derivative at the projection point:

$$d^2 = (e_a - e_b)^2 + 4V_{ab}V_{ba}, \qquad \partial_\Delta d^2 = 2(e_a - e_b)(s_a - s_b). \tag{9}$$

At a nondegenerate minimum of a positive real gap, distinct slopes imply $e_a = e_b$ and hence

$$\boxed{g_N^2 = V_{ab}V_{ba}.} \tag{10}$$

This identity uses the exact finite-size spectral subspace. It organizes the gap but is not an independent prediction of its exponent.

Under $R_i \mapsto c_i R_i$, $W_i \mapsto c_i^{-1} W_i$,

$$V_{ab} \mapsto (c_b/c_a)V_{ab}, \qquad V_{ba} \mapsto (c_a/c_b)V_{ba}. \tag{11}$$

Individual coupling exponents can therefore depend on normalization if c_i depend on N. Switching from population-peak to physical trace-norm normalization changes both exponents while preserving their sum, 2σ. A balanced convention makes each equal to σ by construction [Fig. 3; definitions and values in the SM]. Thus this decomposition alone cannot assign an invariant activation interpretation to a directional exponent.

The weak matrix element contains cancellations involving coherences: at $N = 40$, its output-diagonal and output-off-diagonal contributions cancel by almost four orders of magnitude (SM). This partition does not separate two invariant physical processes. Since $W_b^T(L - cI)R_a = W_b^T L R_a$ for any scalar c, choosing $c = e_a$ transfers the cancellation into the small residual $(L - e_a)R_a$. A right population tail alone therefore does not determine the full biorthogonal matrix element; the boundary-flux analysis in the SM reaches the same limitation.

B. Analytic continuation in detuning

We separately continue the full generator as $L_0 + \zeta L_1$ for complex detuning ζ. This continuation uses $-i[H(\zeta), \rho]$; it is not $-i[H\rho - \rho H^\dagger]$. The continued evolution preserves trace but generally not Hermiticity. It is therefore not a completely positive and trace-preserving (CPTP) physical map.

Full-generator continuation locates nearby complex EPs for $N = 24, 32, 40$. At each point, the projected traceless matrix is nonzero and squares to zero within numerical precision. At $N = 32$, encircling the EP exchanges the two eigenvalues after one circuit and restores them after two (coordinates and stability checks in the SM).

The local linear model predicts

$$|\operatorname{Im}\zeta_{\mathrm{EP}}| \simeq \frac{2g_N}{|s_a - s_b|}, \tag{12}$$

in subpercent agreement with the continued spectra at all three sizes. The complex EP distance thus tracks the real-axis mixing scale for these crossings. This local result does not imply a nearby EP for every avoided crossing or a physical degeneracy at real parameters.

V. EXACT ELIMINATION OF COHERENCE SECTORS

A. Spectral-parameter-dependent effective operator

We use projection methods that retain the spectral parameter [25, 26], evaluated by matrix continued fractions as in spin master equations [27]. Group the Dicke operator basis by signed coherence order $q = m - m'$. The diagonal Hamiltonian and local decay preserve q, whereas the drive connects q only to $q \pm 1$. This gives a block-tridiagonal generator while retaining every Dicke j block.

Let $A_q = L_{q,q}$, $U_q = L_{q,q+1}$, and $D_q = L_{q+1,q}$. Starting at the highest positive order $q_{\mathrm{top}} = N$, define

$$\begin{aligned} &\Sigma_{q_{\mathrm{top}}}(z) = 0, \\ &\Sigma_{q-1}(z) = U_{q-1}[zI - A_q - \Sigma_q(z)]^{-1} D_{q-1}. \end{aligned} \tag{13}$$

Here z is an eigenvalue variable, distinct from the complex detuning ζ above. For real physical parameters and real z, Hermitian conjugation relates the positive and negative coherence chains. Their sum yields

$$K_N(z) = A_0 + 2\operatorname{Re}\Sigma_0(z), \qquad K_N(z)r_0 = z r_0. \tag{14}$$

The $q = 0$ sector still contains the populations of every Dicke block; it is not the excitation-number marginal alone, and $K_N(z)$ need not be a classical Markov generator.

With P projecting onto $q = 0$ and $Q = I - P$, the Schur identity reads

$$\det(zI - L) = \det(zI - L_{QQ}) \det[zI - K_N(z)]. \tag{15}$$

It applies where $z - L_{QQ}$ is invertible; the recursion also requires invertible intermediate denominators. Purely coherent modes with zero $q = 0$ projection and points at elimination poles require separate treatment. For complex z, the analytic continuation involves $\Sigma_+(z) + \overline{\Sigma_+(\bar{z})}$, not $2\operatorname{Re}\Sigma_+(z)$. The complex-detuning EPs above are computed separately from the full generator.

The spectral dependence has a time-domain counterpart in the projection-operator memory formalism [28, 29]. Direct elimination of x_Q from our linear equations gives

$$\begin{aligned} \dot{x}_0(t) = {} & L_{00}x_0(t) + L_{0Q}e^{tL_{QQ}}x_Q(0) \\ & + \int_0^t L_{0Q}e^{(t-s)L_{QQ}}L_{Q0}x_0(s)\,ds. \end{aligned} \tag{16}$$

Thus the self-energy represents memory carried by the eliminated coherence sectors. Replacing it by a constant is an additional approximation.

B. Two roots of one effective branch

In the local real spectral window, select a simple real eigenvalue branch $\mu_N(z)$ of $K_N(z)$ and define $f_N(z) =$

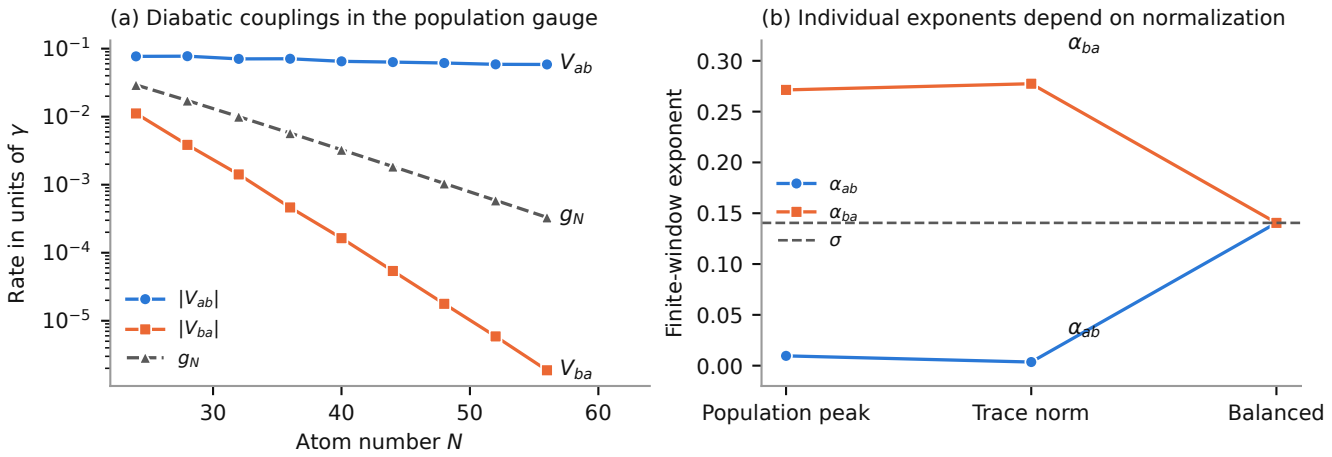


FIG. 3. Biorthogonal gap decomposition. (a) Couplings in the population-peak convention, together with the invariant half-gap. (b) Finite-window exponents under three normalizations of the same two-dimensional subspace. The physical trace-norm convention changes the individual exponents; balancing the couplings makes each equal to σ by construction. The invariant statement is their product and the sum of their logarithmic slopes.

$\mu_N(z) - z$. We find a negative local minimum,

$$f_N(z_c) = -\epsilon_N < 0, \qquad f'_N(z_c) = 0, \\ c_N = f''_N(z_c) > 0. \tag{17}$$

Its two nearby roots give the two decay eigenvalues. Expanding around the minimum gives

$$f_N(z) = -\epsilon_N + \frac{c_N}{2}(z - z_c)^2 + \cdots, \qquad g_{N,\mathrm{quad}} = \sqrt{\frac{2\epsilon_N}{c_N}}. \tag{18}$$

The derivatives are calculated analytically through the matrix recursion and the non-Hermitian effective eigenproblem (SM).

For $N = 24, 32, 40, 48$, direct roots of $f_N = 0$ reproduce the independent full-generator pair; back-substitution and numerical checks are given in the SM. The quadratic approximation has percent-level accuracy at the smallest size and improves rapidly with N [Fig. 4]. Higher-order terms also shift the root midpoint away from z_c.

Self-consistency in z is essential: at $N = 48$, freezing K_N at z_c leaves only one eigenvalue near the full-generator pair (SM). The two roots therefore arise from the spectral dependence of one effective branch, rather than two nearby eigenvalues of this frozen operator. This does not exclude other, correctly matched effective models.

Over the computed window, the minimum depth falls rapidly while the curvature changes modestly [Fig. 4]. Neither trend independently fixes the large-N scaling of ϵ_N/c_N. In particular, no argument establishes an algebraic curvature prefactor. Equation (18) describes the finite-size gap but does not predict its exponent from an action.

C. Connection to biorthogonal self-orthogonality

The effective spectral slope has a precise relation to the full-mode norm. Write $G = (zI - L_{QQ})^{-1}$, $B = L_{0Q}$, and $C = L_{Q0}$, so that $K = L_{00} + BGC$ and $K_z = -BG^2C$. For a simple effective eigenbranch with $Kr = \mu r$, $l^T K = \mu l^T$, and $l^T r = 1$, define the back-substituted vectors

$$R = \begin{pmatrix} r \\ GCr \end{pmatrix}, \qquad W^T = (l^T,\, l^T BG). \tag{19}$$

Then

$$\boxed{W^T R = l^T(I - K_z)r = 1 - \mu_z = -f_z.} \tag{20}$$

At a root, this is the standard derivative pairing for the nonlinear eigenvalue problem $T(z) = zI - K(z)$ [26, 30]. At the negative minimum, $f_z = 0$ but $f \neq 0$: the back-substituted vectors are not eigenmodes, so this is not an EP. Under regular elimination and a simple effective branch, $f = f_z = 0$ with $f_{zz} \neq 0$ would instead give an EP2 (SM). This criterion does not determine the observed depth ϵ_N.

D. Initial coherences and observable residues

The same elimination must retain the initial-coherence source in Eq. (16). For an initial operator $x = (x_0, x_Q)^T$, its reduced Laplace transform obeys

$$[zI - K(z)]\widetilde{x}_0(z) = b(z), \qquad b(z) = x_0 + BG(z)x_Q. \tag{21}$$

For the population observable, $o^T = (o_0^T, 0)$, the standard simple-pole residue [26, 30] becomes

$$a_i = \frac{(o_0^T r_i)[l_i^T b(\lambda_i)]}{l_i^T [I - K'(\lambda_i)] r_i}. \tag{22}$$

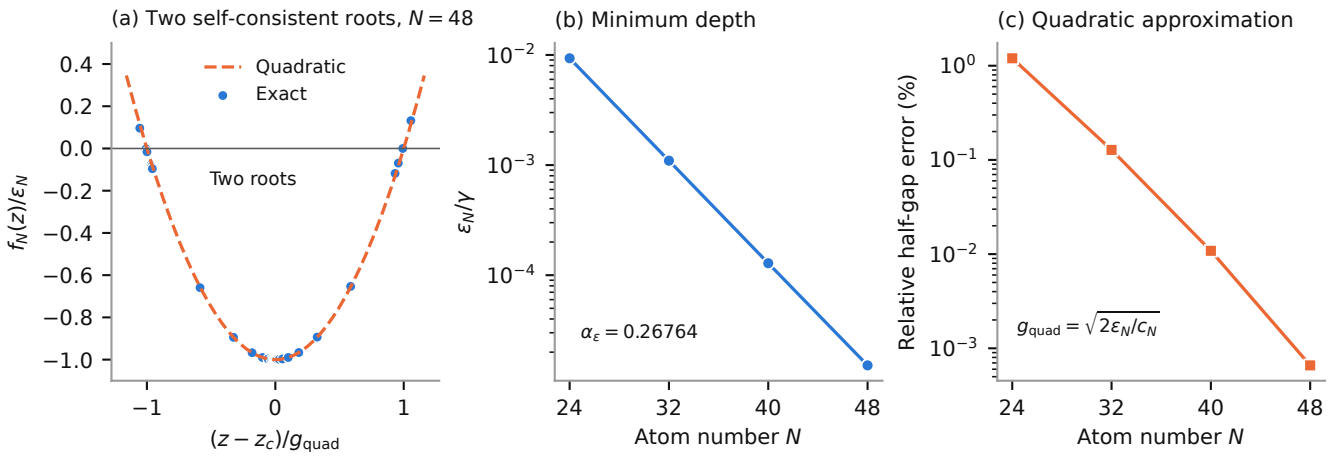


FIG. 4. Exact coherence elimination at the supplied real-axis crossing detunings. (a) The normalized self-consistency function for $N = 48$ (points) and its local quadratic expansion (dashed). (b) Negative-minimum depth and its descriptive four-size exponent. (c) Relative error of the quadratic half-gap. The exact eliminated roots have much smaller errors than this local expansion. Both the depth and curvature are outputs of a finite-N spectral calculation; no independent large-N action is inferred.

Across the four tested sizes, Eq. (22) recovers the full-mode residues. Removing BGx_Q while keeping the same roots and denominator gives order-unity or larger relative errors for the high-branch source (SM). Correct rates alone do not fix observable weights.

A physical control compares the fold product state ρ_f with its globally excitation-number-dephased state $\rho_d = \sum_k \Pi_k \rho_f \Pi_k$. The states share their initial $q = 0$ populations, yet their difference $\delta\rho_c = \rho_f - \rho_d$ generates a nonzero density transient through its coherences; a source depending only on x_0 would predict zero. Yet the two real modes alone retain an order-unity relative error even in the late-time window at the $N = 32$ crossing (SM). Source reconstruction therefore does not establish a two-mode dynamical closure.

VI. DYNAMICAL REACH AND LIMITS OF THE REAL-MODE PAIR

A. Accuracy of spectral truncations

Large expansion coefficients and preparation-dependent visibility require a direct dynamical comparison [10, 11, 21]. We compare the two-real-mode sum with full evolution for the preparations in Sec. III, then include the slower oscillatory conjugate pair at $N = 32$. Sparse exponential multiplication supplies the reference without fitting transient amplitudes or rates.

At the crossing, adding the oscillatory pair reduces the high-branch relative L^2 error over $6 \le t \le 12$ from about 25% to 0.5%. The fold-state four-mode error remains about an order of magnitude larger, demonstrating preparation dependence. The SM gives the full error table and displays spectral contributions after removing the common decay envelope; the reported errors use unscaled signals. These comparisons require the oscillatory pair over the tested window, while the real-pair mean decay rate remains of order γ.

B. Fixed-preparation tests across the crossing

To distinguish mode exchange from changes in preparation, we freeze the high-branch difference at $\Delta_*(N)$, denoting it by $\delta\rho^*_{\mathrm{hi}}$. We also use the coherence difference $\delta\rho_c$, with the same single-atom fold seed at every size and detuning. For each fixed source x, the complete density signal is

$$s_N(t, \Delta) = \mathrm{Tr}\left[\frac{N_e}{N} e^{tL(\Delta)} x\right]. \tag{23}$$

We compare $\Delta_* \pm w_N$, where $w_N = 2g_N/|s_a - s_b|$ is the local mixing scale in Eq. (12). The scans cover $N = 24, 32, 40, 48$ and sample $0 \le t \le 12$, without removing any decay envelope.

For the frozen high-branch source at $N = 48$, individual residues change by order 10^{-3} across this interval, whereas the maximum sampled change of the complete density signal is of order 10^{-6} [Fig. 5(c,d)]. Direct propagation of the first two detuning derivatives accurately predicts both endpoint signals from the center, without a fit or spectral truncation. These comparisons support smooth local response despite strong redistribution of spectral weights. The SM gives derivative errors, coherence-source controls, and two-pole and trace-contraction bounds.

The coherence-source peak at $\Delta_*(32)$ is smaller than its independent-atom counterpart [Fig. 5(a)]. At this

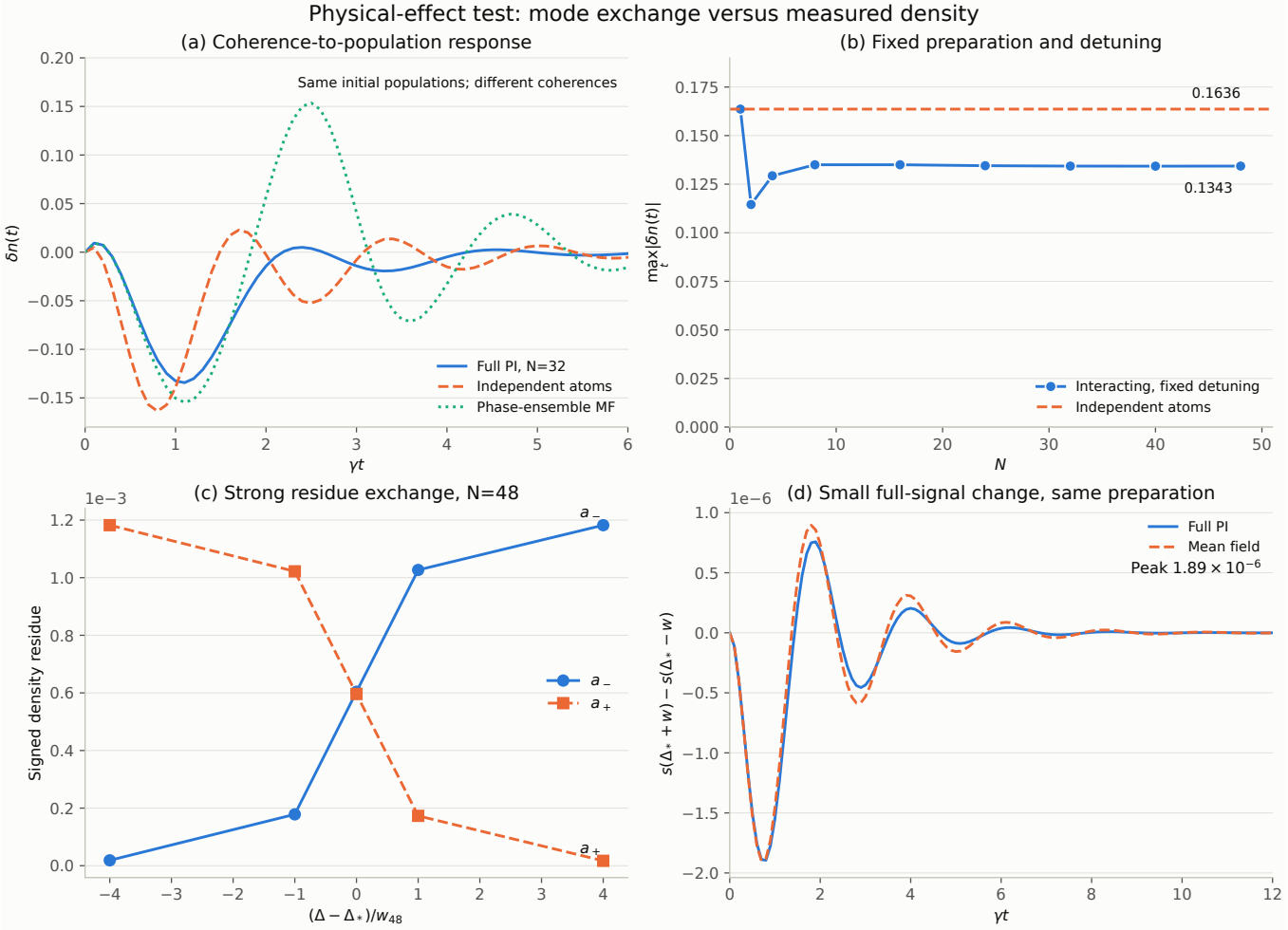


FIG. 5. Fixed-preparation density controls, in raw density units. (a) Fold coherence difference at $N = 32$, $\Delta = \Delta_*(32)$: full permutation-invariant evolution, independent atoms, and a common-phase mean-field ensemble. (b) Largest sampled magnitude at the same detuning and single-atom seed, for $t = 0, 0.1, \ldots, 12$. (c) Signed real-mode residues for the $N = 48$ high-branch difference frozen at its crossing. (d) Difference between the complete signals at $\Delta_* + w_{48}$ and $\Delta_* - w_{48}$ for that fixed preparation. Lines in (b,c) guide the eye; residues in (c) are decomposition weights, whereas (d) compares physical density signals.

fixed detuning and single-atom seed, its magnitude stays near 0.13 over $N = 8$–48 [Fig. 5(b)]. The peak therefore supplies no evidence of enhancement over independent atoms or growth with size in this control. The global initial trace distance varies across sizes (SM).

Mean field nevertheless fails to reproduce the complete coherence-source transient [Fig. 5(a)]. Its control evolves each common-phase product state separately before averaging, retaining the finite-N interaction factor $(N-1)/N$. The discrepancy cannot be assigned to the target pair, whose standalone error for this preparation is already large. The negative enhancement and narrow-response tests apply to the stated preparations, density observable, sizes, and time window; they do not exclude other protocols.

VII. DISCUSSION AND CONCLUSIONS

The locally damped Rydberg ensemble exhibits a finite-size exchange of relaxation residues without a corresponding strong, narrow density response in the tested fixed preparations. This connects mode character, exact coherence elimination, and full transients within one model. Earlier collective-spin studies established finite-rate doublets and cancelling residues [20, 21]; the present calculation resolves their detuning-dependent response counterpart with all local-decay spin blocks retained. The initial-coherence source is essential to that connection, because correct decay roots alone do not determine observable weights.

The independent-atom and fixed-detuning controls restrict the physical interpretation of the coherence-to-population peak. Its amplitude does not identify the

near-degenerate pair, even though the complete transient differs substantially from mean field. Thus the spectral exchange is a statement about the decomposition of relaxation, not evidence for enhanced population switching. Identifying the many-body origin of the remaining mean-field discrepancy requires resolving spectral contributions beyond this pair. Other preparations, observables, or time-dependent protocols remain outside the present tests.

An asymptotic explanation would require the scaling of ϵ_N/c_N, or equivalently $V_{ab}V_{ba}$, independently of the exact pair. Exact steady-state information is available [12], but the barrier conjecture of Noachtar and Clerk [16] addresses the ordinary Liouvillian gap through a specified purification. It does not predict this finite-rate splitting from an arbitrary excitation-number marginal potential. Applying switching-instanton theory [14, 15] likewise requires boundary conditions appropriate to this excited spectral pair. The reported finite-window exponents establish neither derivation nor a thermodynamic fold correspondence.

DATA AND NUMERICAL METHODS

The SM specifies the generator, preparations, projection identities, control calculations, and numerical checks. Plotted records, failed numerical attempts and their refinements, and reproduction scripts accompany the source bundle. Independent full-spin checks cover $N = 1$–4, and detuning derivatives are checked against independent small-system time integration. The finite-size calculations use no fitted stochastic noise model.

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